
INTERNATIONAL GCSE

COMPUTER SCIENCE

Paper 1 Programming

Friday 1 November 2024

07:00 GMT

Time allowed: 2 hours

Materials

For this paper you must have access to:

- a computer
- a printer
- appropriate software
- electronic versions of the **Preliminary Material** and the **Skeleton Program**
- a hard copy of the **Preliminary Material** and the **Skeleton Program**.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 80.
- No extra time is allowed for printing and collating.
- The question paper is divided into **three** sections.
- You may use a bilingual dictionary.
- You must **not** use an English dictionary.
- You are **not** allowed to use a calculator.

Instructions

- Type the information required on the front of your **Electronic Answer Document**.
- Answer **all** questions.
- Enter your answers into the **Electronic Answer Document**.
- Before the start of the examination make sure your **centre number**, **candidate name** and **candidate number** are shown clearly in the footer of every page of your **Electronic Answer Document** (not the front cover).
- Include the evidence required for your answers to **Sections B** and **C** in your **Electronic Answer Document**.
- You **must save** your **Electronic Answer Document** at regular intervals.
- The questions in **Sections B** and **C** require you to make changes to the **Skeleton Program**.
- All of the programming questions in **Sections B** and **C** can be answered independently of each other. If you cannot answer one of the questions you can still attempt to solve later questions.
- You are advised to keep a backup copy of the original **Skeleton Program** so that you can go back to it if you accidentally make changes to the program which means it can no longer be compiled/executed while answering the questions in **Sections B** and **C**.

Secure all your printed Electronic Answer Document pages together and hand them to the invigilator.

Section A (Non-programming questions)

You are advised to spend no more than **30 minutes** on this section.

Type your answers to **Section A** in your **Electronic Answer Document**.

You **must save** your **Electronic Answer Document** at regular intervals.

The questions in this section are about programming and how the **Skeleton Program** works.
Do **not** make any changes to the **Skeleton Program** when answering these questions.

0	1
---	---

.

1

 State the name of a subroutine with two parameters in the **Skeleton Program**.
[1 mark]

0	1
---	---

.

2

 State the name of a local variable from the `displayImage` subroutine.
[1 mark]

0	1
---	---

.

3

 The `loadImage` subroutine has a statement which makes use of concatenation.
Explain what is meant by the term concatenation.
[1 mark]

0	1
---	---

.

4

 The `darkenImage` subroutine makes use of a minimum function.
Explain why this minimum function is needed. You should refer to the `GRAPHICS` constant in your answer.
[3 marks]

0	1
---	---

.

5

 The `main` subroutine asks the user to enter a choice. Any choice entered is then converted to uppercase.
Explain why the value entered is converted to uppercase.
[2 marks]

0	2
---	---

Figure 1 shows the pseudocode for an algorithm that manipulates the pixel data for an image.

Figure 1

```
FOR position  $\leftarrow$  0 TO LENGTH(image) - 1
  IF position MOD 2 = 0 THEN
    image[position]  $\leftarrow$  4 - image[position]
  ENDIF
ENDFOR
```

0	2	.	1
---	---	---	---

Define the term algorithm.

[2 marks]

0	2	.	2
---	---	---	---

Explain why the $- 1$ is needed at the end of the first line of the algorithm in **Figure 1**.

[1 mark]

0	2	.	3
---	---	---	---

Describe the changes necessary to alter the pseudocode so that it uses indefinite iteration.

The algorithm should still behave in the same way.

[3 marks]

0	2	.	4
---	---	---	---

The variable `image` currently holds the data:

1, 2, 3, 4, 4, 3, 2, 1, 0

State the data in `image` after carrying out the algorithm in **Figure 1**.

[2 marks]

Turn over for the next question

Turn over ►

0 3

The subroutine `rleImage` is to be tested by completing a trace table.

The variable `image` currently holds the data:

1, 1, 2, 2, 3, 3, 3, 3, 4

0 3 . 1

Complete the trace table for the subroutine call `rleImage(3, 3, image)`

Copy the contents of all the unshaded cells in **Table 1** into your Electronic Answer Document.

You may not need all the rows in **Table 1**.

Table 1

current	count	position	item	OUTPUT

[6 marks]

0 3 . 2

Explain why the last statement is needed in the `rleImage` subroutine.

[1 mark]

Section B (Short programming questions)

You are advised to spend no more than **45 minutes** on this section.

0	4
---	---

The **Skeleton Program** is to be improved so that it displays the image when the user picks the run length encode image option.

The image should be displayed when the run length encode image option is picked from the menu **before** any of the run length encoded output is shown.

Change the subroutine `rleImage` so that the `displayImage` subroutine is called **before** the run length encoded output is shown.

A blank line must be displayed between the image and the run length encoded output.

Test your changes by running the **Skeleton Program** and then:

- enter 1 to load an image
- enter `one` for the filename of the image
- enter 3 to run length encode the image.

Evidence that you need to provide

Include the following evidence in your Electronic Answer Document.

0	4	.	1
---	---	---	---

All the PROGRAM SOURCE CODE for the amended subroutine `rleImage`.

[3 marks]

0	4	.	2
---	---	---	---

SCREEN CAPTURE(S) showing the results of the requested test.

[1 mark]

Turn over for the next question

Turn over ►

0 5

The **Skeleton Program** is to be improved so that it displays a message about the image.

After the user picks option 1 and loads an image into the program a message about the image dimensions should be displayed as shown in **Figure 2**.

Figure 2

```
1..Load image
2..Display image
3..Run length encode image
4..Stretch image
5..Lighten image
6..Darken image
Q..Quit

Please enter a choice: 1
Please enter the image name: one
The image has size 5 by 5 pixels
```

Change the subroutine `main`.

The message `The image has size X by Y pixels` should be displayed after the image has been loaded. `X` is the width and `Y` is the height of the image.

Test your changes by running the **Skeleton Program** and then:

- enter 1 to load an image
- enter `pattern` for the filename of the image.

Evidence that you need to provide

Include the following evidence in your Electronic Answer Document.

0 5 . 1

All the PROGRAM SOURCE CODE for the amended subroutine `main`.

[4 marks]

0 5 . 2

SCREEN CAPTURE(S) showing the results of the requested test.

[1 mark]

0	6
---	---

The **Skeleton Program** is to be improved so that the stretch factor is validated when the user picks option 4 to stretch the image.

The stretch factor entered should be less than 6.

Change the `stretchImage` subroutine so that:

- when the user enters a value for the stretch factor it is checked to make sure it is less than 6
- if the stretch factor is not less than 6 the program should display the message
`Invalid input - please try again`
- the two previous steps should repeat until the user enters a valid stretch factor.

Test your changes by running the **Skeleton Program** and then:

- enter 1 to load an image
- enter `one` for the filename of the image
- enter 4 to stretch the image
- enter 6 as a stretch factor
- enter 3 as a stretch factor.

Evidence that you need to provide

Include the following evidence in your Electronic Answer Document.

0	6	1
---	---	---

All the PROGRAM SOURCE CODE for the amended subroutine `stretchImage`.

[5 marks]

0	6	2
---	---	---

SCREEN CAPTURE(S) showing the results of the requested test.

[1 mark]

Turn over for the next question

Turn over ►

0	7
---	---

The **Skeleton Program** is to be changed by adding an option to create an image made up of random pixels.

This image will have a width of 5 pixels and a height of 6 pixels.

Each pixel in the random image should have a value that is a random number in the range 0 to 4 including the values 0 and 4

Change the `loadImage` subroutine so that:

- if the user enters `random` as the image name
 - an image of width 5 and height 6 is created
 - the image is filled with random pixel data as described
 - the image is returned
- otherwise, the `loadImage` subroutine should continue to load the image file specified.

Test your changes by running the **Skeleton Program** and then:

- enter 1 to load an image
- enter `random` for the filename of the image
- enter 2 to display the image.

Evidence that you need to provide

Include the following evidence in your Electronic Answer Document.

0	7	.	1
---	---	---	---

All the PROGRAM SOURCE CODE for the amended subroutine `loadImage`.

[8 marks]

0	7	.	2
---	---	---	---

SCREEN CAPTURE(S) showing the results of the requested test.

[1 mark]

0	8
---	---

The **Skeleton Program** is to be modified so that a summary is displayed after the run length encoded data has been displayed.

The summary should include:

- the storage space needed for the original image
- the storage space needed for the run length encoded image.

Change the `rleImage` subroutine so it:

- calculates the storage space needed for the original image using one byte per pixel
- displays the message `The original image needs X bytes` where `X` is the storage space needed
- calculates the storage space needed for the run length encoded image using one byte per run length value and one byte per pixel
- displays the message `The RLE image needs Y bytes` where `Y` is the storage space needed.

Test your changes by running the **Skeleton Program** and then:

- enter 1 to load an image
- enter `zero` for the filename of the image
- enter 3 to run length encode the image.

Evidence that you need to provide

Include the following evidence in your Electronic Answer Document.

0	8	.	1
---	---	---	---

All the PROGRAM SOURCE CODE for the amended subroutine `rleImage`.

[6 marks]

0	8	.	2
---	---	---	---

SCREEN CAPTURE(S) showing the results of the requested test.

[1 mark]

Turn over for the next section

Turn over ►

Section C (Longer programming questions)

You are advised to spend no more than **45 minutes** on this section.

0 9

Images and other files can hide messages. This is known as steganography.

One of the images has a message hidden in it. The value of the first pixel in each row of the image represents a character as shown in **Table 2**.

Table 2

Pixel value	Character represented
0	"c"
1	"k"
2	"l"
3	"o"
4	" " (a space)

The **Skeleton Program** is to be modified so that after displaying an image it will also output the hidden message.

If a message has not been hidden in the image, the output might not have any meaning.

Task 1

Create a new subroutine called `decodeImage` that:

- has parameters `width` and `image`
- creates a new variable for the hidden message
- examines the value of the first pixel in each row of the image:
 - uses the information in **Table 2** to find the character represented by the value
 - adds the character to the hidden message
- after all pixels have been processed, outputs the hidden message.

Task 2

Modify the `main` subroutine so that, when menu option 2 is selected, the `decodeImage` subroutine is called after the `displayImage` subroutine is called.

Task 3

Test your changes by running the **Skeleton Program** and then:

- enter 1 to load an image
- enter `pattern` for the filename of the image
- enter 2 to display the image.

Evidence that you need to provide

Include the following evidence in your Electronic Answer Document.

0 9 . 1 All the PROGRAM SOURCE CODE for the new subroutine `decodeImage`.
[10 marks]

0 9 . 2 All the PROGRAM SOURCE CODE for the amended subroutine `main`.
[1 mark]

0 9 . 3 SCREEN CAPTURE(S) showing the results of the requested test.
[1 mark]

Turn over for the next question

Turn over ►

1 0

The **Skeleton Program** is to be modified so that an image can be extended.

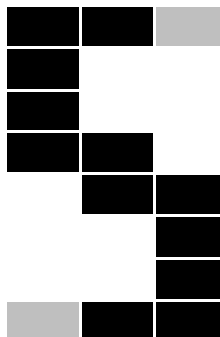
The extended image includes a rotated version of the original image below the original image, as shown in **Figure 3**.

Figure 3

Original image



Extended image



In **Figure 3**, the original image data is:

4, 4, 2, 4, 0, 0, 4, 0, 0, 4, 4, 0

In **Figure 3**, the extended image data is:

4, 4, 2, 4, 0, 0, 4, 0, 0, 4, 4, 0, 0, 4, 4, 0, 0, 4, 0, 0, 4, 2, 4, 4

The extended image data contains the original pixel data followed by the original pixel data reversed.

Task 1

Create a new subroutine called `extendImage` that:

- has parameters `width`, `height` and `image`
- creates a blank new image by calling the `createImage` subroutine with the correct values for the width and height of the **extended** image
- sets the pixel data for the new image to the original image pixel data followed by the original image pixel data reversed
- returns the new image and the new height.

Task 2

Modify the `displayMenu` subroutine so that 7 is displayed as the option for extending the image.

Task 3

Modify the `main` subroutine so that when 7 is entered at the main menu prompt:

- the `extendImage` subroutine is called
- **and**
- the values returned are stored in the `image` and `height` variables.

Task 4

Test your changes by running the **Skeleton Program** and then:

- enter 1 to load an image
- enter `one` for the filename of the image
- enter 7 to extend the image
- enter 2 to display the image.

Evidence that you need to provide

Include the following evidence in your Electronic Answer Document.

1	0
---	---

.

1

 All the PROGRAM SOURCE CODE for the new subroutine `extendImage`.
[9 marks]

1	0
---	---

.

2

 All the PROGRAM SOURCE CODE for the amended subroutine `displayMenu`.
[1 mark]

1	0
---	---

.

3

 All the PROGRAM SOURCE CODE for the amended subroutine `main`.
[3 marks]

1	0
---	---

.

4

 SCREEN CAPTURE(S) showing the results of the requested test.
[1 mark]

END OF QUESTIONS

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